EXP:4
4/2

VERIFICATION OF THEVENIN'S THEOREM

AIM:

To verify the thevenin's theorem Pratically and theorectically for the given Dc circuit.

APPARATUS REQUIRED

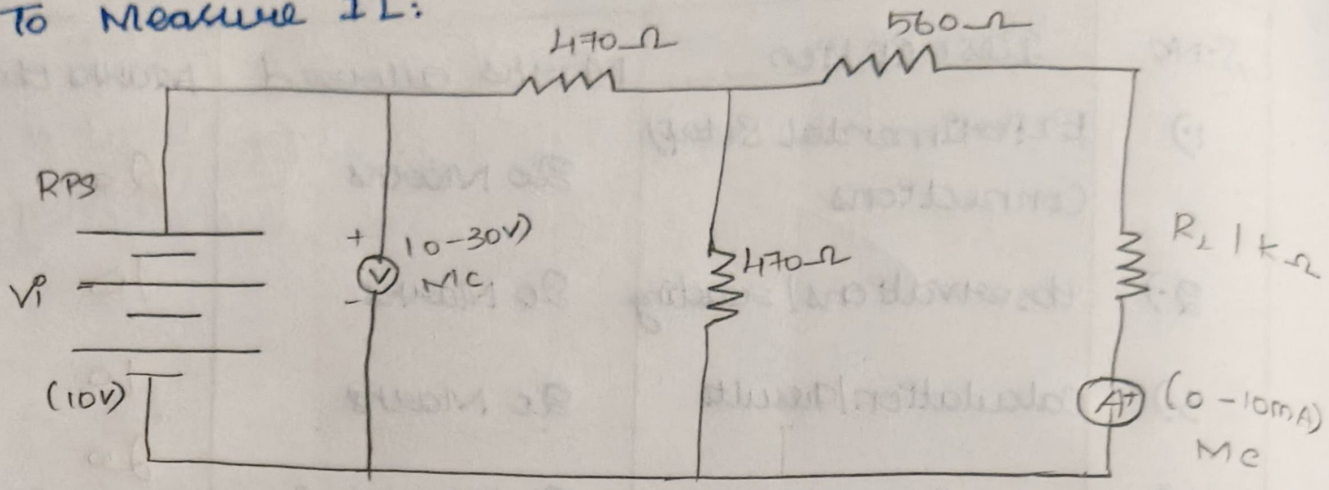
S.NO	APPARATUS	SPECIFICATION	QUANTITY
1.	Regulated Power supply	(0-30V)	1
2.	Voltmeter	(0-30)MC	1
3.	Ammeter	(0-10mA) MC	1
4.	Register's	470Ω, 560Ω 3KΩ	2
5.	Bread Board	-	1
6.	multimeter	-	1

PROCEDURE:

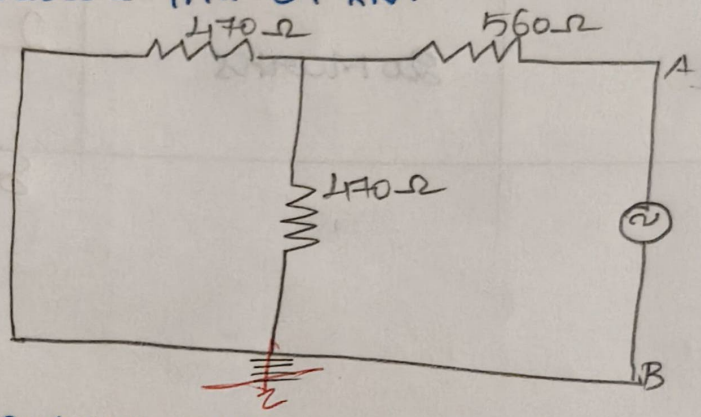
1.) Make the connection as per circuit digram
2.) vary the RPS and get on input voltage of 10V
3.) Note down the voltmeter reading (V) and ammeter reading (I) in Tabular column,
4.) Switch off the supply and make connections for digram 2
5.) Measure thevenin's resistance $R_{th} = R_N$
6.) Switch off the supply and make connections for

Circuit Diagram

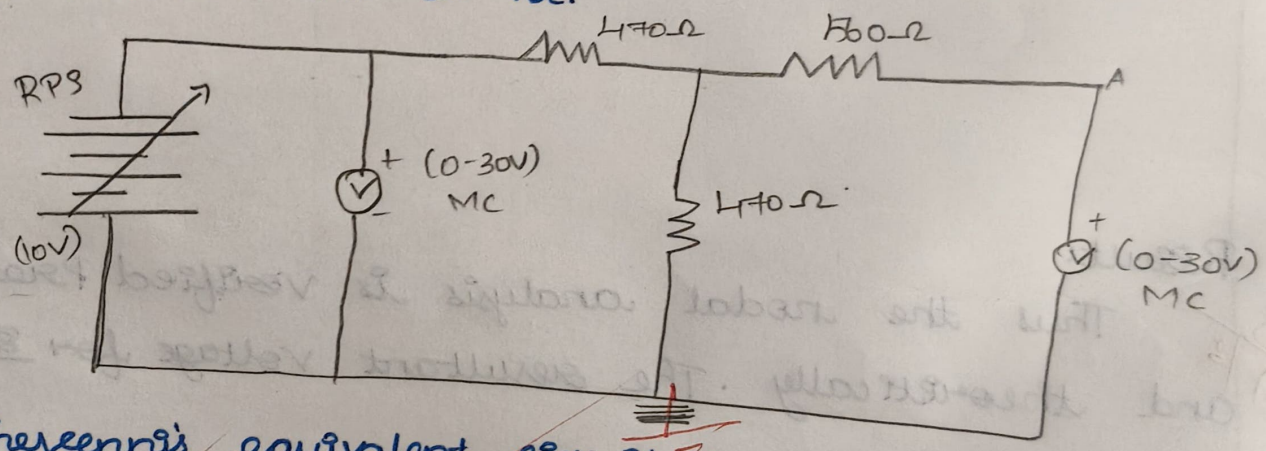
To Measure IL:



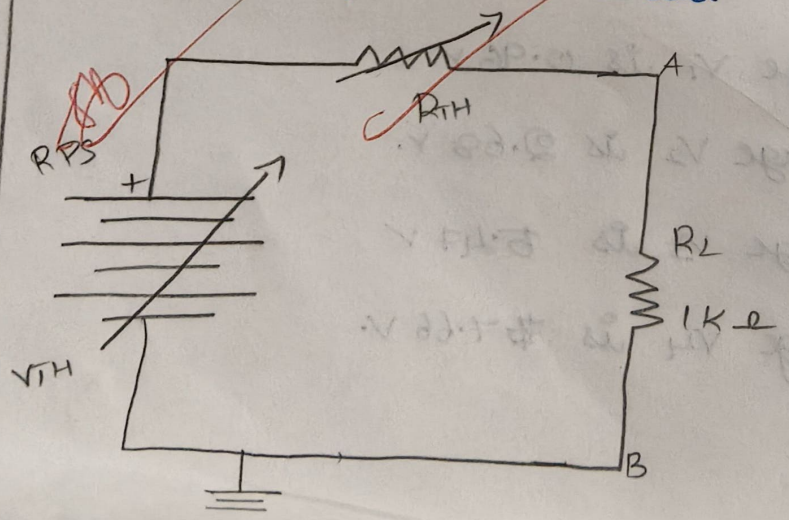
To Measure RTH or RN:



To Measure VTH or VOC:



Thevenin's equivalent circuit



Tabular column:

To Measure I_L

V_1 (Volts)	I_L (amps)
2	0.59
4	1.186
6	1.73

Tabular column: 2

To Measure R_{TH} or R_N

From the circuit diagram $2R_{th} = R_N = 798\ \Omega$

To Measure V_{TH} or V_{oc}

V_1 (Volts)	V_{TH} (Volts)
2	1.06
4	2.09
6	3.10

8/10

dp required

* Set an input voltage of 10V in the RPS and note down the voltmeter reading V_1 and V_{Th} = (= Voc) in

Tabular column: 3

$$I_L = \frac{V_{TH}}{R_{TN} + R_L}$$

$$I_L = \frac{I_N \times R_N}{R_N + R_L}$$

Model Calculation

Practical Value of I_L (from tabulation D) = 2.3 MA

Verification of Thevenin's theorem:

$I_L = V_{TH} / (R_{TH} + R_L) = 2.22 mA$

Theoretical calculation of I_L, R_{TH} (R_N) and V_{TH} for the given circuit

For calculating N_{TH}:

$$R_1 = \frac{470 \times 470}{470 + 470} = 235\,\Omega$$

$$V_{TH} = V_L \times \frac{R_2}{R_1 + R_2} = \frac{10 \times 235}{470 + 235} = 10 \times \frac{235}{705}$$

$$= 10 \times 0.333$$

$$V_{TH} = 0.333V$$

Marks Split up

S.No	Description	Marks allowed	Marks obtained
1.)	Experimental Setup/ Connection	20 marks	20
2.)	observation/reading	20 marks	10
3.)	calculations/result	20 marks	10
4.)	Viva	20 marks	20
5.)	Record	20 marks	20
			80

80

17/10

Result:

Thus thevenin's theorem is verified Pratically and theoretically.